



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO3-AT2 Data Interpretation

Slot A

DSA 0404 – Fundamentals of Data Science

K. Lahari

192325118

AIML

1. Sampling Distribution Reasoning

Question: A population has mean $\mu = 500$ and SD $\sigma = 80$. A random sample of $n = 64$ is drawn. (a) Compute the standard error of the sample mean. (b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal? (c) If n were increased to 256, how does the standard error change, and what principle explains this?

(a) Standard Error

Step 1: Formula

$$SE = \sigma / \sqrt{n}$$

Step 2: Substitute values

$$SE = 80 / \sqrt{64} = 80 / 8$$

Step 3: Result

$$SE = 10$$

(b) Probability sample mean exceeds 520

Step 1: Formula

$$Z = (\bar{x} - \mu) / SE$$

Step 2: Substitute values

$$\begin{aligned} Z &= (520 - 500) / 10 \\ &= 20 / 10 \\ &= 2 \end{aligned}$$

Step 3: Look up standard normal table

$$\begin{aligned} P(Z < 2) &= 0.9772 \\ P(Z > 2) &= 1 - 0.9772 \\ &= 0.0228 \end{aligned}$$

Step 6: Result

$$\text{Probability} \approx 0.0228 \text{ (2.28\%)}$$

(c) Effect of increasing n to 256

Step 1: Formula

$$SE = \sigma / \sqrt{n}$$

Step 2: Substitute values

$$\begin{aligned} SE &= 80 / \sqrt{256} \\ &= 80 / 16 \end{aligned}$$

$$= 5$$

Step 3: Compare

SE dropped from 10 to 5 (halved), while n increased 4×

Step 4: Principle

$SE \propto 1/\sqrt{n}$, so precision increases with the square root of sample size, not linearly with n

2. Covariance Calculation and Sign Interpretation

Question: Given paired data on advertising spend (X, ₹'000) and sales (Y, ₹ lakh): X: 10, 20, 30, 40, 50; Y: 15, 18, 25, 30, 42. Calculate Cov(X,Y). Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

Step 1: Compute means

$$\bar{X} = (10+20+30+40+50)/5 = 30,$$

$$\bar{Y} = (15+18+25+30+42)/5 = 26$$

Step 2: Compute deviations from mean

$$X - \bar{X}: -20, -10, 0, 10, 20$$

$$Y - \bar{Y}: -11, -8, -1, 4, 16$$

Step 3: Multiply each pair of deviations

$$(-20)(-11) = 220,$$

$$(-10)(-8) = 80,$$

$$(0)(-1) = 0,$$

$$(10)(4) = 40,$$

$$(20)(16) = 320$$

Step 4: Sum the products

$$220+80+0+40+320$$

$$= 660$$

Step 5: Divide by n

$$\text{Cov}(X,Y) = 660 / 5$$

$$= 132$$

Step 6: Interpret sign

$\text{Cov}(X,Y)$ is positive, so as X increases, Y increases too — consistent with intuition (higher ad spend, higher sales)

Step 7: Limitation

The magnitude 132 is unit-dependent ($\text{₹'000} \times \text{₹ lakh}$) and unbounded, so it cannot show the strength of the relationship. Correlation $r = \text{Cov}(X,Y)/(\sigma_x \cdot \sigma_y)$ standardizes this to a fixed -1 to $+1$ scale, revealing strength as well as direction

3. Correlation vs Causation Trap

Question: A city reports that ice cream sales and drowning incidents have a correlation coefficient of $r = 0.85$. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

Step 1: Identify the flaw

- $r = 0.85$ only shows association between the two variables, not that one causes the other. This is the correlation-implies-causation fallacy.

Step 2: Identify the confounder

- Temperature/season (hot weather). Heat independently drives up both ice cream sales and swimming activity, which raises drowning risk.

Step 3: Explain the mechanism

- Because a single common cause (heat) affects both variables, they rise and fall together without either one causing the other.

Step 4: List evidence needed to establish real causation

- Controlled experiment manipulating ice cream consumption directly
- Temporal precedence (cause consistently occurs before effect)
- Statistical control for temperature/season — correlation should disappear if it was the confounder
- A plausible biological/physical mechanism (none exists here)
- A dose-response relationship
- Replication of the finding across cities/years

Step 5: Conclusion

Without this evidence, $r = 0.85$ reflects a shared seasonal cause, not causation.

4. Effect of Outliers on Correlation

Question: A dataset of 9 points shows $r = 0.42$ (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

Step 1: Compare the two r values

- r went from 0.42 to 0.89 after removing just 1 point out of 9 (more than 11% of the data).

Step 2: Explain why this happens

- Pearson's r is computed from actual numeric deviations from the mean, so one point far from the rest can dominate the covariance and variance terms used in the formula.

Step 3: Conclusion on fragility

- this shows Pearson's r is not robust; a single outlier in a small sample can seriously distort the calculated relationship.

Step 4: Consider Spearman's rank correlation

- Spearman converts values to ranks before correlating, so an outlier only occupies one rank position (e.g., highest or lowest) instead of exerting influence through its numeric distance from the mean.

Step 5: Justify robustness

- Because rank position is bounded, a single extreme value has limited effect on Spearman's r . Here, Spearman's r would likely stay close to 0.89 whether or not the outlier is included, making it the more stable choice for this small, outlier-affected dataset.

5. Data Preparation — Missing Values

Question: A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents (not random). (a) Classify this missing data mechanism (MCAR/MAR/MNAR). (b) Why would mean imputation introduce bias here? (c) Suggest a better strategy.

(a) Classification

Step 1: Check if missingness relates to any variable

- yes, it is linked to employment status, so it is not MCAR.

Step 2: Check whether that variable is observed

- employment status is recorded/observed in the dataset.

Step 3: Apply the definition

- since missingness depends on an observed variable (employment status) rather than directly on the hidden income value itself, this is MAR (Missing At Random).

Step 4: Note the edge case

- If missingness were driven purely by the unrecorded income value itself, independent of employment status being known, it would instead be MNAR.

(b) Why mean imputation is biased

Step 1: Identify who is missing

- Mostly unemployed respondents, who likely have lower or zero income.

Step 2: Identify what mean imputation does

- It fills every missing value with the overall sample mean, computed mainly from employed respondents' incomes.

Step 3: Identify the resulting bias

- This overestimates income for the unemployed subgroup, pulling overall statistics upward.

Step 4: Identify the secondary problem

- It also artificially shrinks the variance of Income, since all imputed values collapse to one single number.

(c) Better strategy

Step 1: Use the observed relationship

- Since missingness depends on employment status, condition the imputation on that variable.

Step 2: Options

- Stratified imputation: fill missing values using the mean/median income within each employment-status subgroup
- Regression imputation: predict Income from employment status, education, age, etc., and use predicted values
- Multiple Imputation (e.g., MICE): generate several imputed datasets conditioned on observed variables, analyze each, then pool results

Step 3: Result

- Any of these preserves the true relationship between Income and employment status and avoids the bias and variance-shrinkage caused by simple mean imputation.